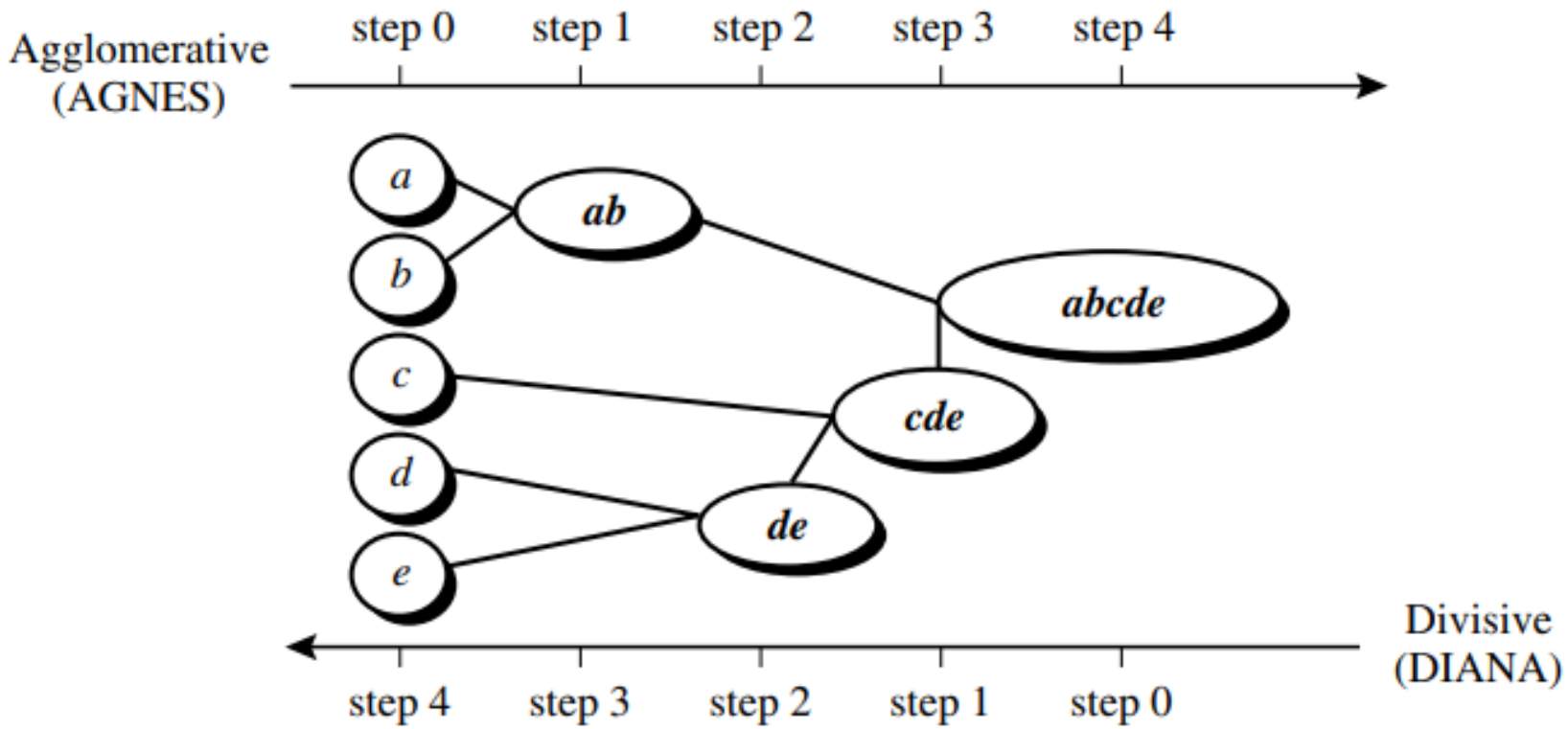
Hierarchical clustering

Hierarchical Clustering

- **Hierarchical methods:** A hierarchical method creates a hierarchical decomposition of the given set of data objects.
- **Hierarchical cluster analysis** or **HCA** is an unsupervised clustering algorithm which involves creating clusters that have predominant ordering from top to bottom.
- Strategies for hierarchical clustering generally fall into two categories:
  - **Agglomerative**: This is a "bottom-up" approach: Each observation starts in its own cluster, and pairs of clusters are merged as one moves up the hierarchy.
  - **Divisive**: This is a "top-down" approach: All observations start in one cluster, and splits are performed recursively as one moves down the hierarchy.

Dendrogram: A tree structure called a dendrogram is commonly used to represent the process of hierarchical clustering. It shows how objects are grouped together step by step.

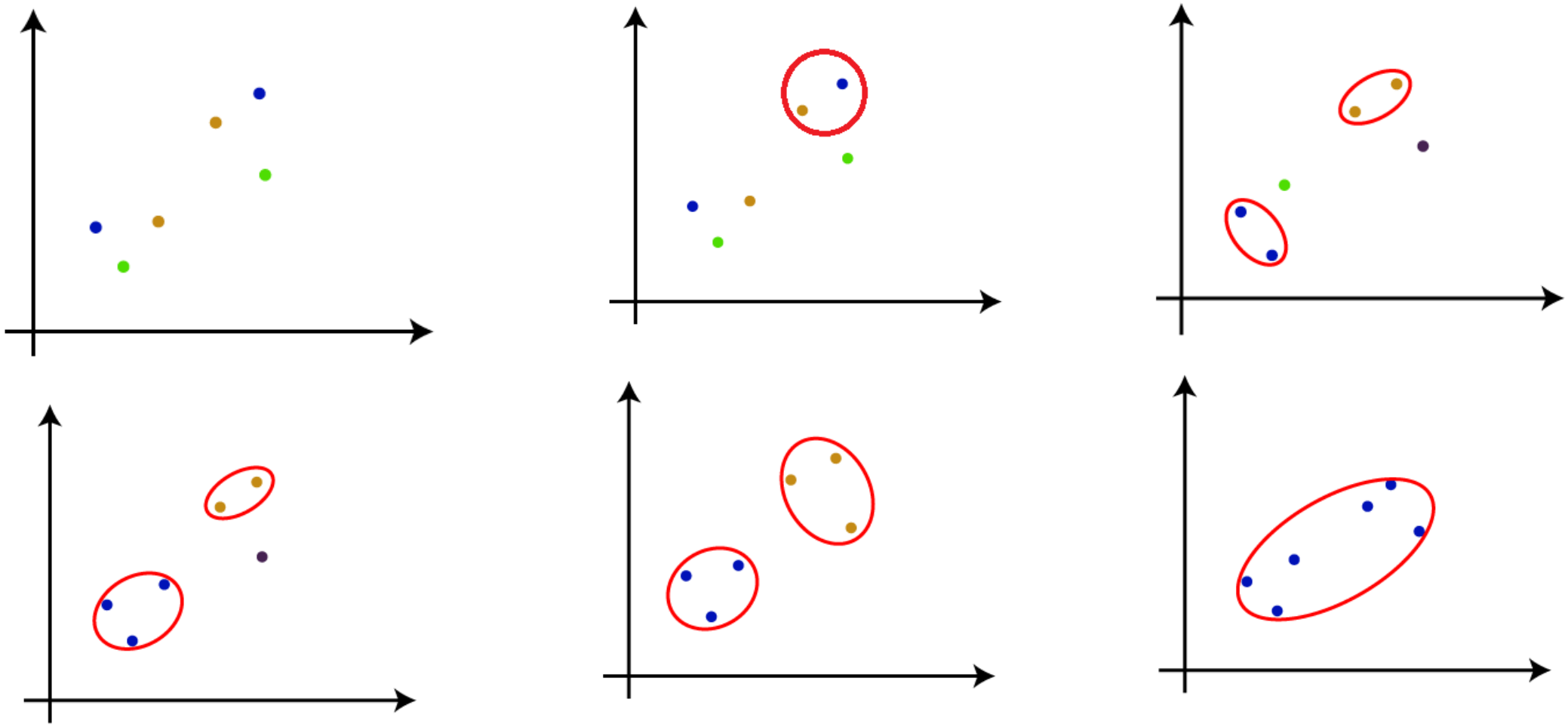


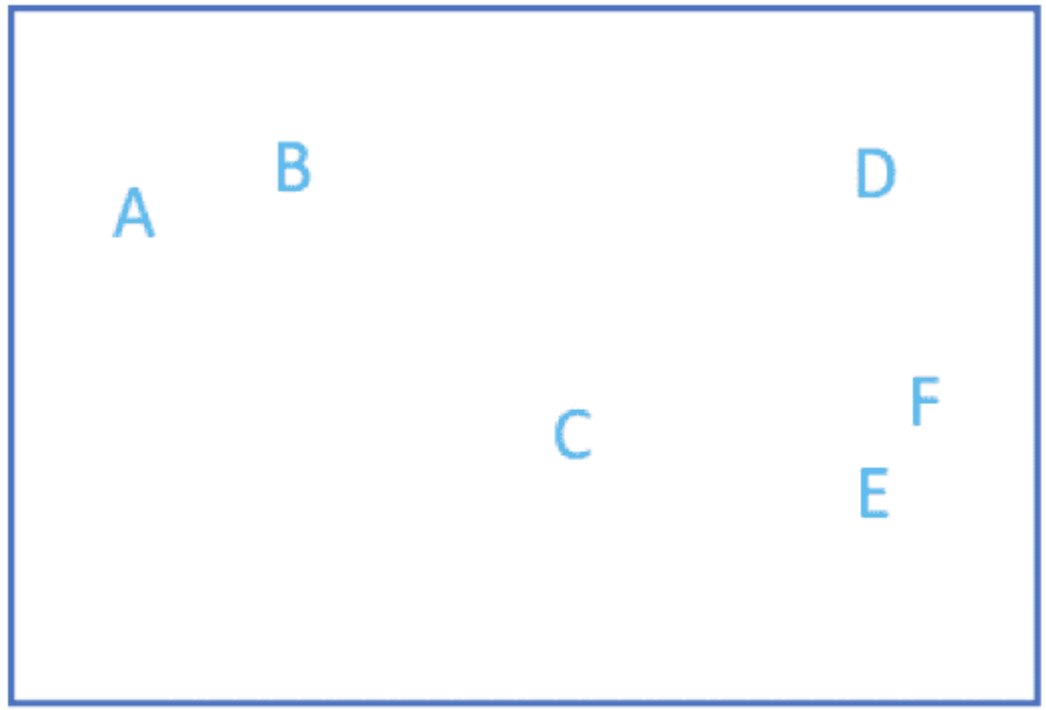
7.6 Agglomerative and divisive hierarchical clustering on data objects $\{a, b, c, d, e\}$.

Agglomerative Hierarchical Clustering

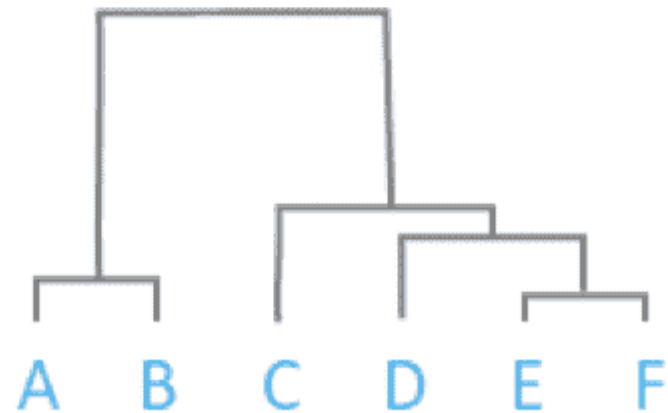
- The Agglomerative Hierarchical Clustering is the most common type of hierarchical clustering used to group objects in clusters based on their similarity.
- It's also known as AGNES (Agglomerative Nesting). It's a "bottom-up" approach
- Each observation starts in its own cluster forming a separate group.
- It successively merges the objects or groups that are close to one another, until all of the groups are merged into one (the topmost level of the hierarchy), or until a termination condition holds.
- **How does it work?**
  1. Initially consider each data point a single-point individual cluster → forms N clusters
  2. Calculate the similarity/ Proximity of one cluster with all the other clusters
  3. Merge the clusters which are highly similar or close to each other.
  4. Recalculate the proximity matrix for each cluster
  5. Repeat step 3 and 4 until only a single cluster remains.

➢ Agglomerative HC Illustration





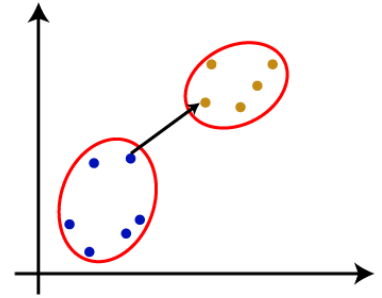
Dendrogram



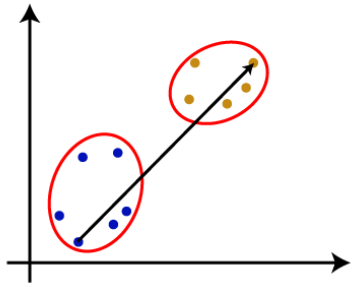
Measures of distance in HC (similarity)

- The choice of distance metric should be made based on theoretical concerns from the domain of study.
- That is, a distance metric needs to define similarity in a way that is sensible for the field of study.
  - For example, if clustering crime sites in a city, city block distance may be appropriate. Or, better yet, the time taken to travel between each location.
  - Where there is no theoretical justification for an alternative, the Euclidean should generally be preferred, as it is usually the appropriate measure of distance in the physical world.
- Linkage Criteria
  - After selecting a distance metric, it is necessary to determine linkage for merging the clusters.
  - linkage refers to how we measure the distance between clusters (groups of points) in hierarchical clustering.
  - Linkage measures “How far apart are two clusters?”

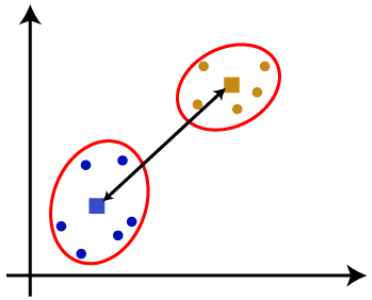
- Single-Linkage
  - Single-linkage (nearest neighbor) is the shortest distance between a pair of observations in two clusters.
  - It can sometimes produce clusters where observations in different clusters are closer together than to observations within their own clusters.
  - Tends to form elongated or chain-like clusters.
  - Sensitive to noise and outliers



- Complete-Linkage
  - Complete-linkage (farthest neighbor) is where distance is measured between the farthest pair of observations in two clusters.
  - Produces compact and well-separated clusters.
  - Less sensitive to noise, but may break large clusters.



- Average-Linkage (Unweighted Pair Group Mean Averaging)
  - Distance between clusters = average of all pairwise distances between points in the two clusters.
  - Provides a balance between single and complete linkage.
  - More stable and less affected by extreme values.



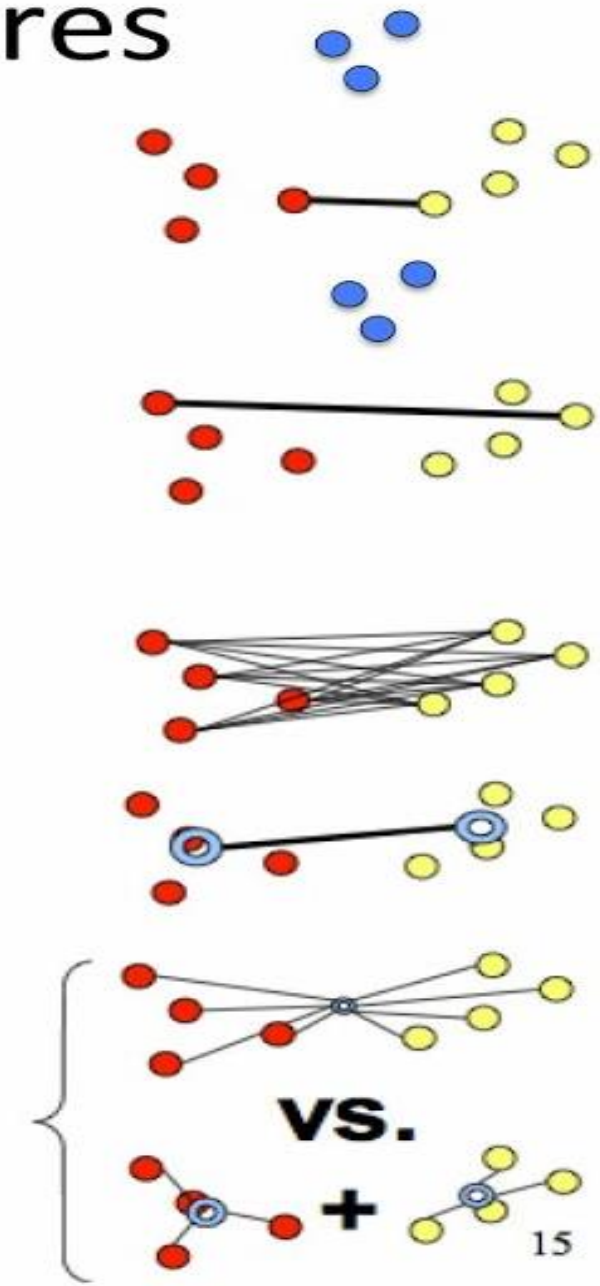
- Centroid-Linkage
  - Centroid-linkage is the distance between the centroids of two clusters.
  - As the centroids move with new observations, it is possible that the smaller clusters are more similar to the new larger cluster than to their individual clusters causing an inversion in the dendrogram.
  - This problem doesn’t arise in the other linkage methods because the clusters being merged will always be more similar to themselves than to the new larger cluster.
- Ward’s Linkage
  - Merges clusters that minimize the increase in total within-cluster variance.
  - Produces compact, spherical clusters.
  - Often preferred for quantitative data and Euclidean distance.

🔹 Ward's Linkage (Ward's Method)

- **Idea:**
  At each step, merge the two clusters that cause the **smallest increase in within-cluster variance** (i.e., keeps clusters as homogeneous as possible).
- **Distance Measure:**
  Instead of direct point-to-point distance, it uses the increase in **Sum of Squared Errors (SSE)**:
  - Chooses the merge that **minimizes the total variance increase** after merging.
- **Behavior:**
  - Produces **compact, spherical, and equally sized clusters**.
  - Avoids chaining effect (unlike single linkage).
  - Works best with **Euclidean distance**.
- **Key Advantage:**
  - Often gives **high-quality clustering** when clusters are well-separated.
- **Limitation:**
  - Sensitive to **outliers** (since variance is affected).
  - Not suitable for non-spherical cluster shapes.

Cluster distance measures

- Single link: $D(c_1, c_2) = \min_{x_1 \in c_1, x_2 \in c_2} D(x_1, x_2)$
  - distance between closest elements in clusters
  - produces long chains a→b→c→…→z
- Complete link: $D(c_1, c_2) = \max_{x_1 \in c_1, x_2 \in c_2} D(x_1, x_2)$
  - distance between farthest elements in clusters
  - forces "spherical" clusters with consistent "diameter"
- Average link: $D(c_1, c_2) = \frac{1}{|c_1|}\frac{1}{|c_2|}\sum_{x_1 \in c_1}\sum_{x_2 \in c_2} D(x_1, x_2)$
  - average of all pairwise distances
  - less affected by outliers
- Centroids: $D(c_1, c_2) = D\left(\left(\frac{1}{|c_1|}\sum_{x \in c_1} \vec{x}\right), \left(\frac{1}{|c_2|}\sum_{x \in c_2} \vec{x}\right)\right)$
  - distance between centroids (means) of two clusters
- Ward's method: $TD_{c_1 \cup c_2} = \sum_{x \in c_1 \cup c_2} D(x, \mu_{c_1 \cup c_2})^2$
  - consider joining two clusters, how does it change the total distance (TD) from centroids?



Agglomerative Hierarchical Clustering Solved Example

- Consider the following set of 6 one dimensional data points:
- 18, 22, 25, 42, 27, 43
- Apply the **agglomerative hierarchical clustering** algorithm to build the hierarchical clustering **dendogram**.
- Merge the clusters using **Min distance** and update the proximity matrix accordingly.
- Clearly show the **proximity matrix** corresponding to each iteration of the algorithm.

Compute the proximity matrix

Step – 1

	18	22	25	27	42	43
18	0	4	7	9	24	25
22	4	0	3	5	20	21
25	7	3	0	2	17	18
27	9	5	2	0	15	16
42	24	20	17	15	0	1
43	25	21	18	16	1	0

	18	22	25	27	42	43
18	0	4	7	9	24	25
22	4	0	3	5	20	21
25	7	3	0	2	17	18
27	9	5	2	0	15	16
42	24	20	17	15	0	1
43	25	21	18	16	1	0

(42, 43)

Step – 2

	18	22	25	27	42, 43
18	0	4	7	9	24
22	4	0	3	5	20
25	7	3	0	2	17
27	9	5	2	0	15
42, 43	24	20	17	15	0

Find the clusters using single link technique. Use Euclidean distance, and draw the dendrogram.

	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

- Calculate Euclidean distance, create the distance matrix.

$$\text{Distance } [(x,y), (a,b)] = \sqrt{(x-a)^2+(x-b)^2}$$

$$\text{Distance (P1,P2)} = \sqrt{(0.40-0.22)^2+(0.53-0.38)^2}$$

$$\text{(0.40,0.53), (0.22,0.38)} = \sqrt{(0.18)^2+(0.15)^2}$$

$$= \sqrt{0.0324+0.0225}$$

$$= \sqrt{0.0549}$$

$$= 0.23$$

- The distance matrix is

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

- The distance matrix is

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

- The distance matrix is

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

- To update the distance matrix MIN[dist(P3,P6),P1)]
- MIN(dist(P3,P1), (P6,P1))

  = min[(0.22,0.23)]

  = 0.22
- To update the distance matrix MIN[dist(P3,P6),P2)]
- MIN(dist(P3,P2), (P6,P2))

  = min[(0.15,0.25)]

  = 0.15

- The updated distance matrix for cluster P3, P6

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.22	0.15	0		
P4	0.37	0.20	0.15	0	
P5	0.34	0.14	0.28	0.29	0

- To update the distance matrix MIN[dist(P3,P6),P4)]
- MIN(dist(P3,P4), (P6,P4))

  = min[(0.15,0.22)]

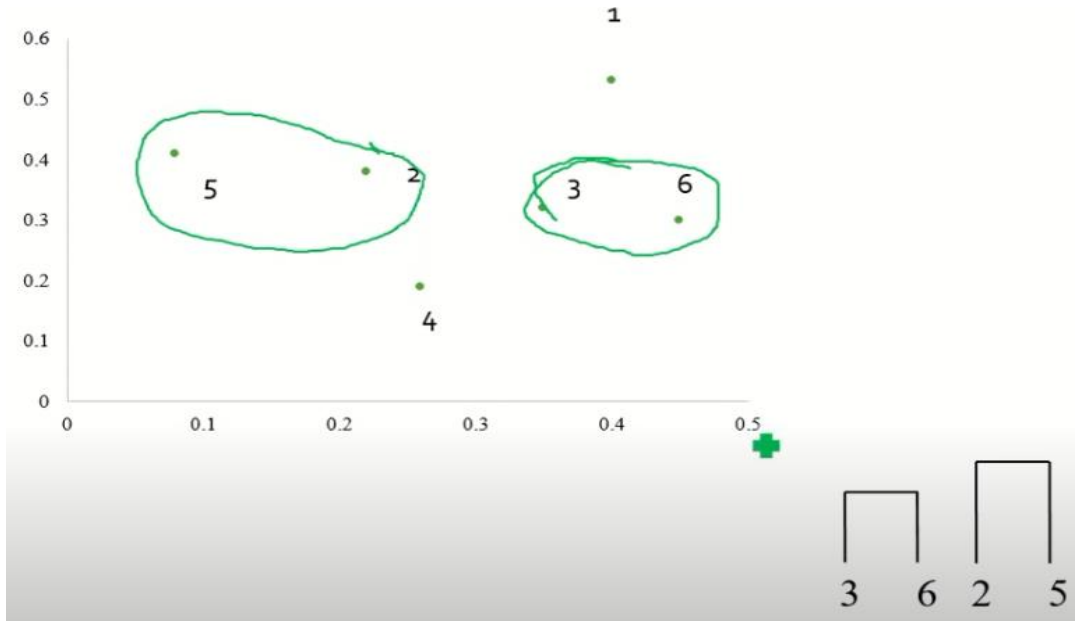
  = 0.15
- To update the distance matrix MIN[dist(P3,P6),P5)]
- MIN(dist(P3,P5), (P6,P5))

  = min[(0.28,0.39)]

  = 0.28

- The distance matrix is

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.22	0.15	0		
P4	0.37	0.20	0.15	0	
P5	0.34	0.14	0.28	0.29	0



- To update the distance matrix MIN[dist(P2,P5),P4)]
- MIN[dist(P2,P4), (P5,P4)]

  = min[(0.20,0.29)]

  = 0.20

- To update the distance matrix MIN[dist(P2,P5),P1)]
- MIN[dist(P2,P1), (P5,P1)]

  = min[(0.23,0.34)]

  = 0.23

- To update the distance matrix MIN[dist(P2,P5),(P3,P6)]
- MIN[dist(P2,(P3,P6)), (P5,(P3,P6))]

  = min[(0.15,0.28)]

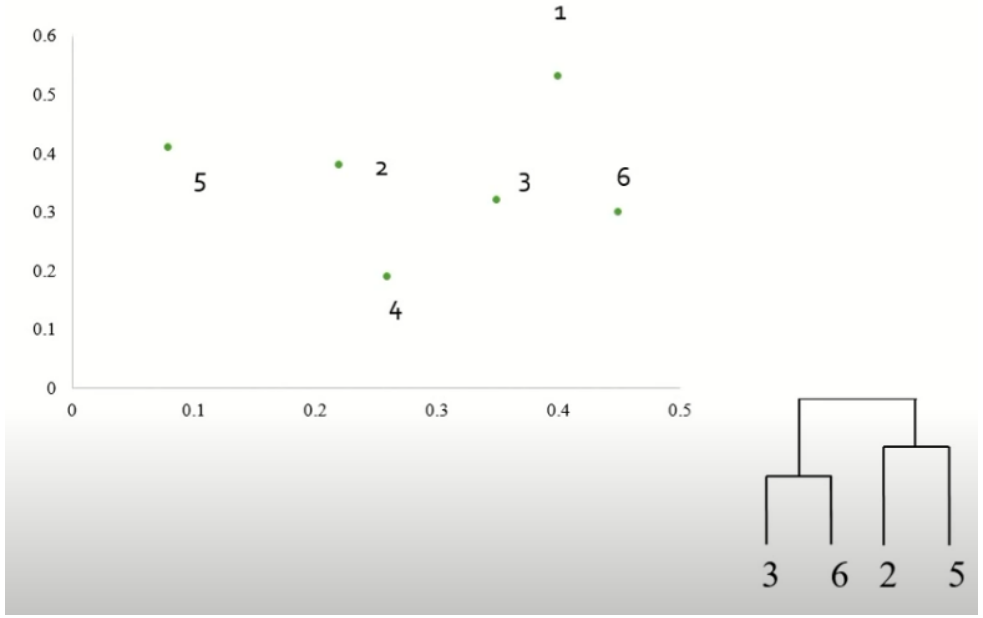
  = 0.15

- The updated distance matrix for cluster P2,P5

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.23	0		
P3,P6	0.22	0.15	0	
P4	0.37	0.20	0.15	0

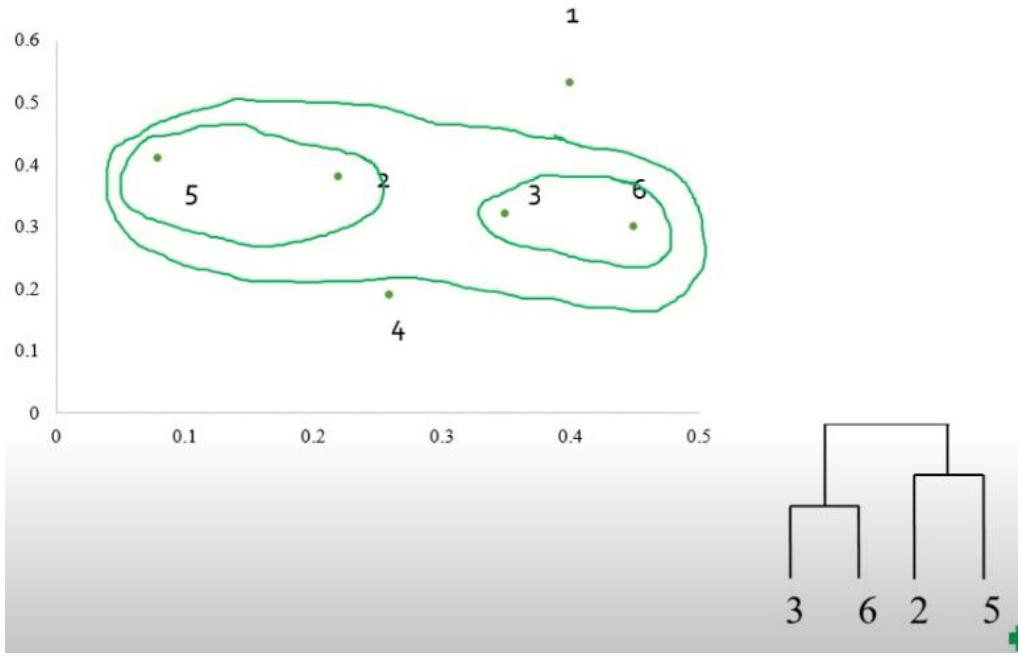
- The distance matrix is

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.23	0		
P3,P6	0.22	0.15	0	
P4	0.37	0.20	0.15	0



- The distance matrix for cluster (P3,P6), (P2,P5)

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.23	0		
P3,P6	0.22	0.15	0	
P4	0.37	0.20	0.15	0



- To update the distance matrix MIN[dist((P2,P5),(P3,P6)),P1]
- MIN[dist((P2,P5),P1), ((P3,P6),P1)]

  = min[(0.23,0.22)]

  = 0.22

- To update the distance matrix MIN[dist((P2,P5),(P3,P6)),P4]
- MIN[dist((P2,P5),P4), ((P3,P6),P4)]

  = min[(0.20,0.15)]

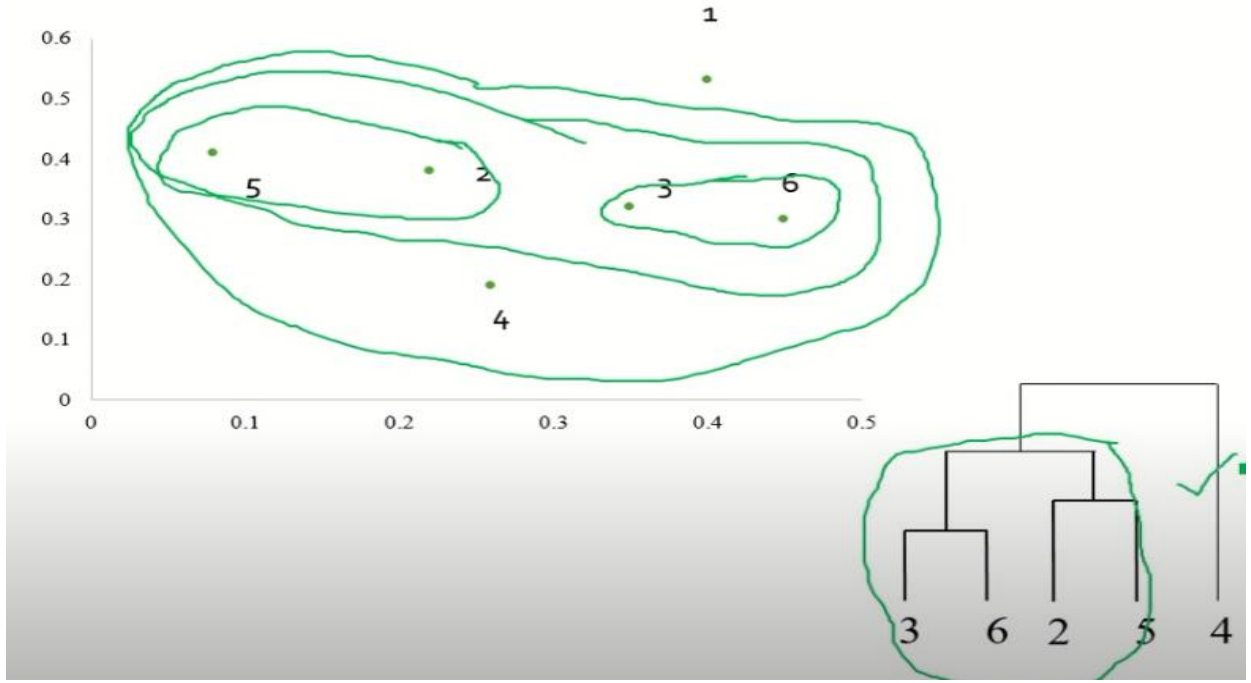
  = 0.15

- The updated distance matrix for cluster P2,P5,P3,P6

	P1	P2,P5,P3,P6	P4
P1	0		
P2,P5,P3,P6	0.22	0	
P4	0.37	0.15	0

- The distance matrix is

	P1	P2,P5,P3,P6	P4
P1	0		
P2,P5,P3,P6	0.22	0	
P4	0.37	0.15	0



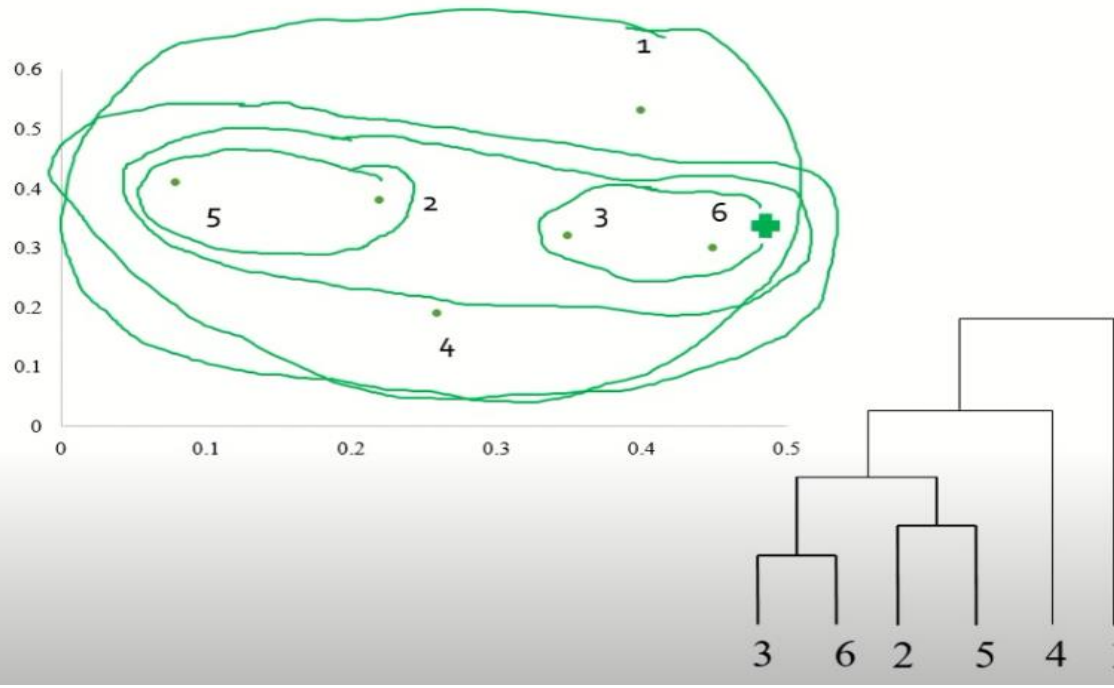
- To update the distance matrix MIN[dist(P2,P5,P3,P6),P4]
- MIN[dist((P2,P5,P3,P6),P1), (P4,P1)]

  = min[(0.22,0.37)]

  = 0.22

- The updated distance matrix for cluster P2,P5,P3,P6,P4

	P1	P2,P5,P3,P6,P4
P1	0	
P2,P5,P3,P6,P4	0.22	0



Complete linkage

Consider 4 points on a line:

- A = 1
- B = 2
- C = 5
- D = 8

Step 1: Compute pairwise distances

- d(A,B) = |1 − 2| = 1
- d(A,C) = 4
- d(A,D) = 7
- d(B,C) = 3
- d(B,D) = 6
- d(C,D) = 3

👉 Smallest distance = **1**, so merge **A and B → Cluster {A,B}**

Step 2: Compute distance using complete linkage

Now clusters: {A,B}, C, D

- d({A,B}, C) = max(d(A,C), d(B,C)) = max(4, 3) = 4
- d({A,B}, D) = max(7, 6) = 7
- d(C, D) = 3

👉 Smallest = **3**, so merge **C and D → Cluster {C,D}**

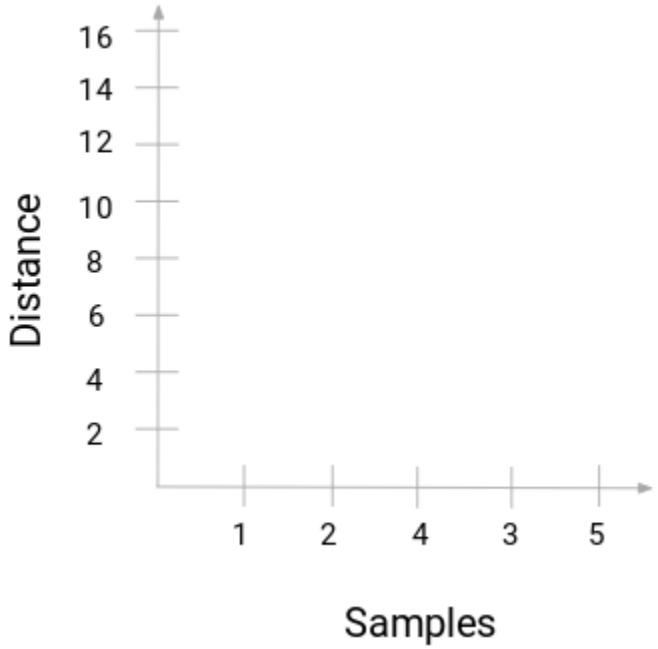
Step 3: Final merge

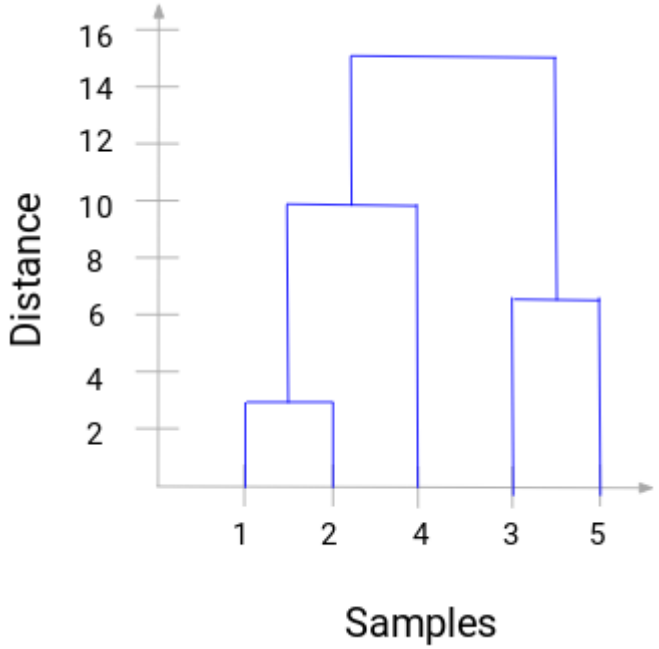
Now clusters: {A,B} and {C,D}

- d({A,B}, {C,D}) = max(4, 7, 3, 6) = **7**

👉 Merge them at distance **7**

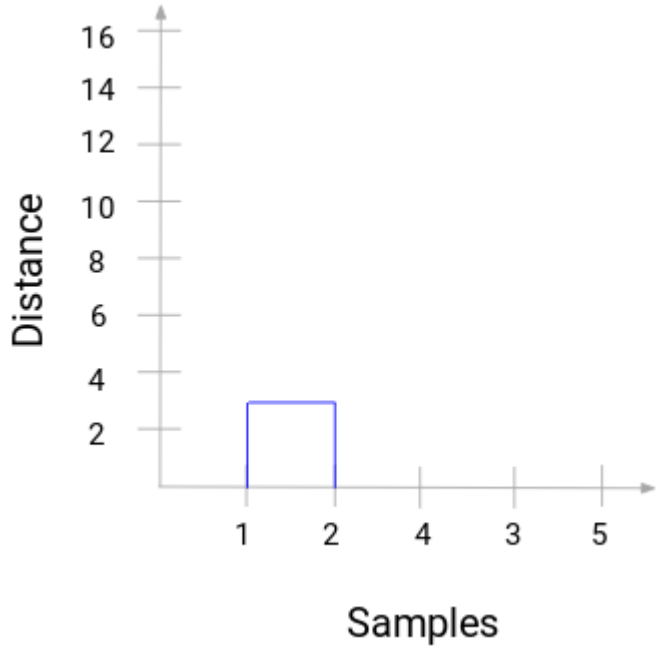
- **How should we choose the number of clusters in hierarchical clustering?**
  - To get the number of clusters for hierarchical clustering, we make use of **dendrogram**.

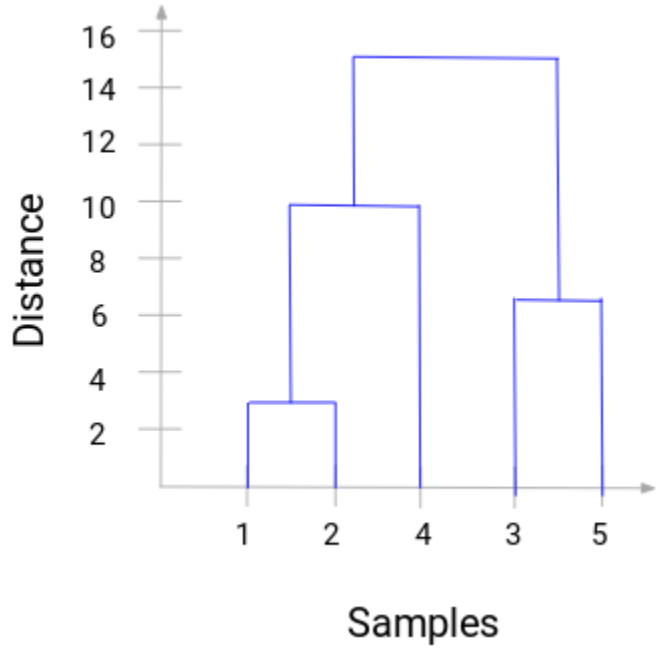




  - Whenever two clusters are merged, we will join them in this dendrogram and the height of the join will be the distance between these points.

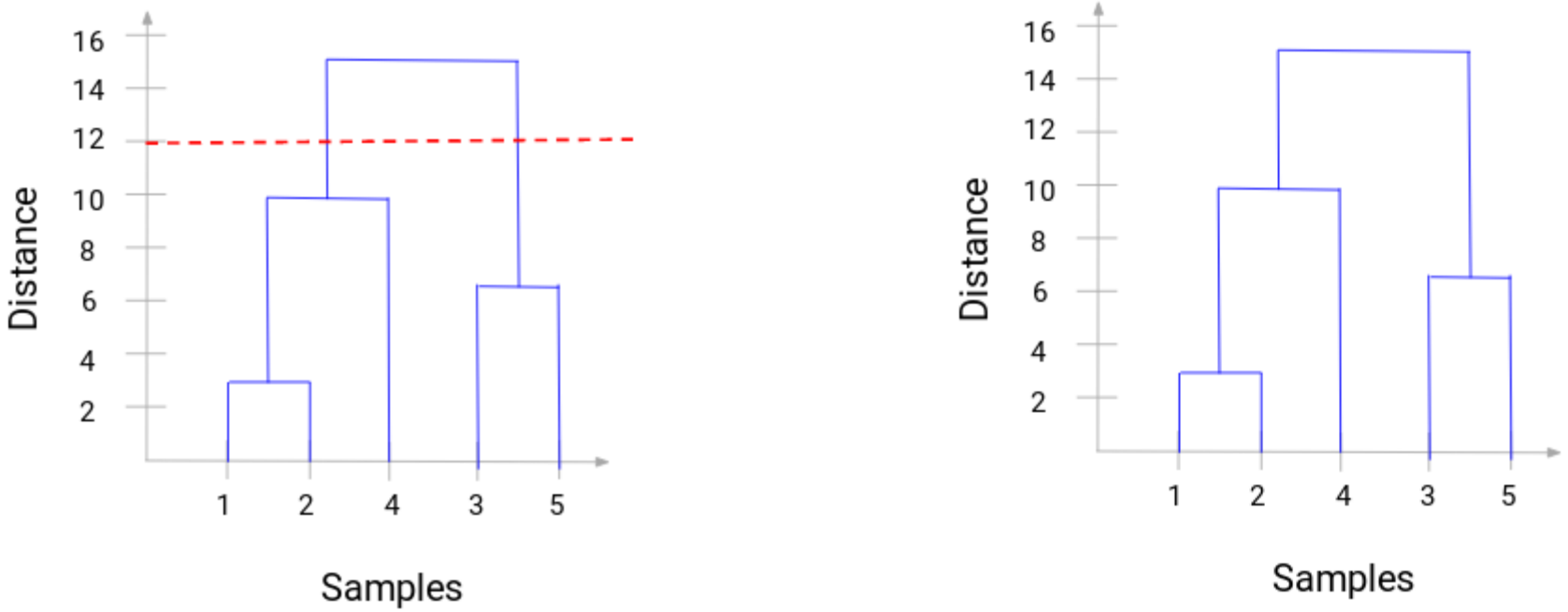
- Example: Assume sample 1 and 2 are merged and the distance between these two samples is 3





- The vertical line represents the distance between these samples.
- More the distance of the vertical lines in the dendrogram, more the distance between those clusters.

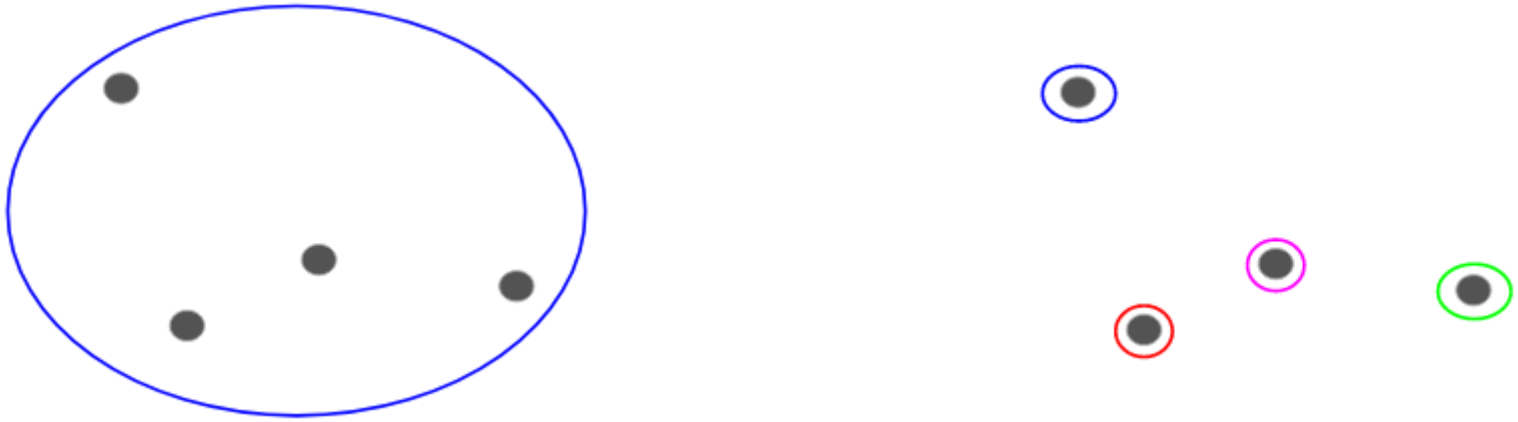
- Now, we can set a threshold distance and draw a horizontal line. Let this threshold be 12 and draw a horizontal line:



- The number of clusters will be the number of vertical lines which are being intersected by the line drawn using the threshold.
- In the above example, since the red line intersects 2 vertical lines, we will have 2 clusters. One cluster will have a sample (1,2,4) and the other will have a sample (3,5).

2. Divisive Clustering(**DI**visive **ANA**lysis- DIANA):

- It is the top-down approach form of hierarchical clustering where all data points are initially assigned a single cluster.
- Further, the clusters are split into two least similar clusters. This is done recursively until clusters groups are formed which are distinct to each other.
- In the end, we are left with N clusters; each cluster only contains a single point.



DIANA is a divisive hierarchical clustering technique. Here is an outline of the algorithm.

Step 1. Suppose that cluster C_l is going to be split into clusters C_i and C_j.

Step 2. Let $C_i = C_l$ and $C_j = \varnothing$.

Step 3. For each object $x \in C_i$:

(a) For the first iteration, compute the average distance of x to all other objects.

(b) For the remaining iterations, compute

$$D_x = \text{average}\,\{d(x,y) : y \in C_i\} - \text{average}\{d(x,y) : y \in C_j\}.$$

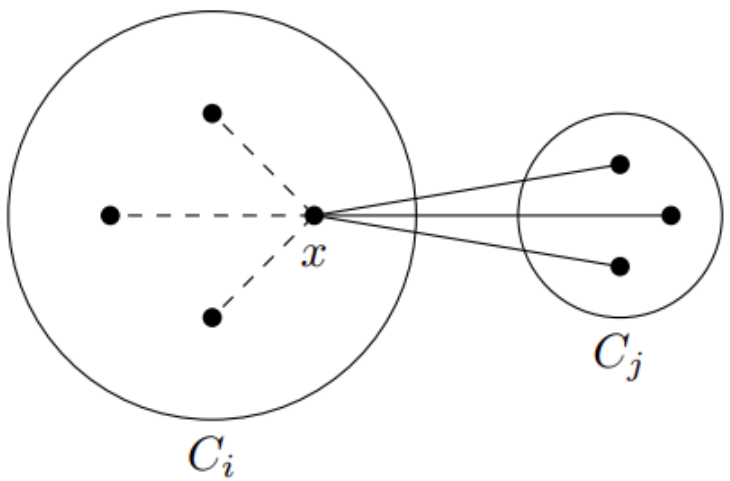


Figure 13.16: D_x= (average of dashed lines) − (average of solid lines)

Step 4. (a) For the first iteration, move the object with the maximum average distance to C_j.

(b) For the remaining iterations, find an object x in C_i for which D_x is the largest. If $D_x > 0$ then move x to C_j.

Step 5. Repeat Steps 3(b) and 4(b) until all differences D_x are negative. Then C_l is split into C_i and C_j.

Step 6. Select the smaller cluster with the largest diameter. (The diameter of a cluster is the largest dissimilarity between any two of its objects.) Then divide this cluster, following Steps 1-5.

Step 7. Repeat Step 6 until all clusters contain only a single object.

Example

- Given the dataset {a, b, c, d, e} and the given distance matrix, construct a dendrogram by the divisive analysis algorithm.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

ole 13.4: Example for distance matrix

Solution

1. We have, initially

$$C_l = \{a, b, c, d, e\}$$

2. We write

$$C_i = C_l, \quad C_j = \varnothing.$$

3. Division into clusters

(a) Initial iteration

Let us calculate the average dissimilarities of the objects in C_i with the other objects in C_i.

Average dissimilarity of a

$$= \frac{1}{4}(d(a,b) + d(a,c) + d(a,e)) = \frac{1}{4}(9 + 3 + 6 + 11) = 7.25$$

Similarly we have :

Average dissimilarity of $b = 7.75$

Average dissimilarity of $c = 5.25$

Average dissimilarity of $d = 7.00$

Average dissimilarity of $e = 7.75$

The highest average distance is 7.75 and there are two corresponding objects. We choose one of them, b, arbitrarily. We move b to C_j.

We now have

$$C_i = \{a, c, d, e\}, \quad C_j = \varnothing \cup \{b\} = \{b\}.$$

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

ple 13.4: Example for distance matrix

(b) Remaining iterations

(i) 2-nd iteration.

$$D_a = \frac{1}{3}(d(a,c) + d(a,d) + d(a,e)) - \frac{1}{1}(d(a,b)) = \frac{20}{3} - 9 = -2.33$$
$$D_c = \frac{1}{3}(d(c,a) + d(c,d) + d(c,e)) - \frac{1}{1}(d(c,b)) = \frac{14}{3} - 7 = -2.33$$
$$D_d = \frac{1}{3}(d(d,a) + d(d,c) + d(d,e)) - \frac{1}{1}(d(c,b)) = \frac{23}{3} - 7 = 0.67$$
$$D_e = \frac{1}{3}(d(e,a) + d(e,c) + d(e,d)) - \frac{1}{1}(d(e,b)) = \frac{21}{3} - 7 = 0$$

D_d is the largest and $D_d > 0$. So we move, d to C_j.
We now have

$$C_i = \{a, c, e\}, \quad C_j = \{b\} \cup \{d\} = \{b, d\}.$$

(ii) 3-rd iteration

$$D_a = \frac{1}{2}(d(a,c) + d(a,e)) - \frac{1}{2}(d(a,b) + d(a,d)) = \frac{14}{2} - \frac{15}{2} = -0.5$$
$$D_c = \frac{1}{2}(d(c,a) + d(c,e)) - \frac{1}{2}(d(c,b) + d(c,d)) = \frac{5}{2} - \frac{16}{2} = -13.5$$
$$D_e = \frac{1}{2}(d(e,a) + d(e,c)) - \frac{1}{2}(d(e,b) + d(e,d)) = \frac{13}{2} - \frac{18}{2} = -2.5$$

All are negative. So we stop and form the clusters C_i and C_j.

4. To divide, C_i and C_j, we compute their diameters.

$$\begin{aligned}\text{diameter}(C_i) &= \max\{d(a,c), d(a,e), d(c,e)\} \\ &= \max\{3, 11, 2\} \\ &= 11 \\ \text{diameter}(C_j) &= \max\{d(b,d)\} \\ &= 5\end{aligned}$$

The cluster with the largest diameter is C_i. So we now split C_i.

We repeat the process by taking C_I = {a, c, e}. The remaining computations are left as an exercise.

Reference

- https://www.youtube.com/watch?v=-yxgNZJjODA
- https://www.analyticsvidhya.com/blog/2019/05/beginners-guide-hierarchical-clustering/